

# MEA OCPP 1.6 Compliance Test Report

Section 4: Reset Check

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-11 12:07

## Contents

|          |                                       |          |
|----------|---------------------------------------|----------|
| <b>1</b> | <b>Test Configuration</b>             | <b>2</b> |
| <b>2</b> | <b>Section 4 Results</b>              | <b>2</b> |
| <b>3</b> | <b>Notes on Failures and Warnings</b> | <b>3</b> |
| <b>4</b> | <b>Dependency Map</b>                 | <b>4</b> |

# 1 Test Configuration

|                   |                                                    |
|-------------------|----------------------------------------------------|
| Device Under Test | Vector vSECC.single Board                          |
| Device IP         | 192.168.1.166                                      |
| OCPP Version      | 1.6 (JSON)                                         |
| CP ID             | rddQC4000001                                       |
| CSMS              | wss://ocpp.measandbox.com:2930                     |
| Connection        | Direct (no proxy)                                  |
| Test PC           | 192.168.111.185 (alias 192.168.1.200/24 on enp3s0) |
| EV Plug Timeout   | 0 s per plug event                                 |
| Boot Wait Timeout | 90 s (after Hard Reset)                            |
| Test Date         | 2026-06-11 04:33:39 UTC                            |
| Results file      | tex/vsecc_section4_results.json                    |

## Test Architecture

Reset commands are issued via the MEA REST API (POST `/EV/remote/reset`). Three reset scenarios are tested:

1. **Hard Reset while idle (4.1–4.8):** Hard Reset is sent when the charger is idle. The vSECC reboots and sends a new BootNotification. A charging session is then started via RemoteStart.
2. **Soft Reset during charging (4.9–4.17):** Soft Reset is sent during an active charging session. The vSECC should gracefully send StopTransaction before rebooting. A new charging session is started after recovery.
3. **Hard Reset during charging (4.18–4.21):** Hard Reset is sent during an active charging session. The session may be aborted immediately; the vSECC reboots and reports Available after reconnection.

# 2 Section 4 Results

Summary: 6 PASS    0 FAIL    15 WARN    0 SKIP    (21 total)

| Item                                                         | Test Description                               | Detail / Evidence                                                                                                         | Result |
|--------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|
| <b>Flow 1: Hard Reset (Idle) — Items 4.1–4.8</b>             |                                                |                                                                                                                           |        |
| 4.1                                                          | Reset (Hard) → Accepted                        | HTTP 200                                                                                                                  | PASS   |
| 4.2                                                          | BootNotification → Accepted (after Hard Reset) | CSMS reconnection confirmed after reset                                                                                   | PASS   |
| 4.3                                                          | StatusNotification Available (after reboot)    | Available not observed on MQTT in 20 s after reset                                                                        | WARN   |
| 4.4                                                          | StatusNotification Preparing (EV plug, Flow 1) | EV_WAIT_SEC=0   skipped<br><i>Connect physical EV to connector and rerun</i>                                              | WARN   |
| 4.5                                                          | RemoteStartTransaction → Accepted              | HTTP 200                                                                                                                  | PASS   |
| 4.6                                                          | StartTransaction (triggered by RemoteStart)    | charging_session_state not observed on MQTT in 30 s                                                                       | WARN   |
| 4.7                                                          | StatusNotification Charging (Flow 1)           | Charging status not observed on MQTT in 20 s                                                                              | WARN   |
| 4.8                                                          | MeterValues (Flow 1, during charging)          | MeterValues not observed on MQTT in 15 s                                                                                  | WARN   |
| <b>Flow 2: Soft Reset (During Charging) — Items 4.9–4.17</b> |                                                |                                                                                                                           |        |
| 4.9                                                          | Reset (Soft) → Accepted (during charging)      | HTTP 200                                                                                                                  | PASS   |
| 4.10                                                         | StopTransaction (triggered by Soft Reset)      | session_state idle/finished not observed in 30 s<br><i>Soft Reset should cause graceful StopTransaction before reboot</i> | WARN   |

| Item                                                          | Test Description                                           | Detail / Evidence                                                                                                            | Result |
|---------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------|
| 4.11                                                          | StatusNotification Finishing (Soft Reset, EV plugged)      | Finishing not observed on MQTT in 15 s<br><i>Some implementations skip Finishing → go directly to Available</i>              | WARN   |
| 4.12                                                          | StatusNotification Available (unplug after Soft Reset)     | Available not observed on MQTT in 30 s   unplug EV                                                                           | WARN   |
| 4.13                                                          | StatusNotification Preparing (EV plug, Flow 2)             | EV_WAIT_SEC=0   skipped<br><i>Connect physical EV to connector and rerun</i>                                                 | WARN   |
| 4.14                                                          | RemoteStartTransaction → Accepted (Flow 2)                 | HTTP 200                                                                                                                     | PASS   |
| 4.15                                                          | StartTransaction (Flow 2, triggered by RemoteStart)        | charging_session_state not observed on MQTT in 30 s                                                                          | WARN   |
| 4.16                                                          | StatusNotification Charging (Flow 2)                       | Charging status not observed on MQTT in 20 s                                                                                 | WARN   |
| 4.17                                                          | MeterValues (Flow 2, during charging)                      | MeterValues not observed on MQTT in 15 s                                                                                     | WARN   |
| <b>Flow 3: Hard Reset (During Charging) — Items 4.18–4.21</b> |                                                            |                                                                                                                              |        |
| 4.18                                                          | Reset (Hard) → Accepted (during charging)                  | HTTP 200                                                                                                                     | PASS   |
| 4.19                                                          | StopTransaction (triggered by Hard Reset)                  | session_state idle/finished not observed in 30 s<br><i>Hard Reset may not send StopTransaction before rebooting</i>          | WARN   |
| 4.20                                                          | StatusNotification Finishing (Hard Reset, EV plugged)      | Finishing not observed on MQTT in 15 s<br><i>Hard Reset typically skips Finishing — goes directly Available after reboot</i> | WARN   |
| 4.21                                                          | StatusNotification Available (after Hard Reset, EV unplug) | Available not observed in 90s   unplug EV / wait for reboot                                                                  | WARN   |

### 3 Notes on Failures and Warnings

#### Reset command

The MEA sandbox REST API endpoint `POST /EV/remote/reset` accepts `"reset_type": "Hard"` or `"reset_type": "Soft"`. Items 4.1, 4.9, and 4.18 are PASS if the API returns HTTP 200.

#### Hard Reset reboot timing

After a Hard Reset the vSECC physically reboots. The test waits up to `BOOT_WAIT_SEC` seconds for the MQTT `vsecc/ocpp_connection.status=connected` event or a successful REST API probe to confirm reconnection.

#### EV-dependent items

- **4.4, 4.13** StatusNotification Preparing: require a physical EV to plug in within the `EV_WAIT_SEC` timeout.
- **4.6–4.8, 4.15–4.17** Charging session items depend on RemoteStart succeeding with an EV present.

#### StopTransaction on reset

- **4.10** Soft Reset: the OCPP specification requires the charger to send StopTransaction with `reason=SoftReset` before restarting.
- **4.19** Hard Reset: the specification allows the charger to immediately reboot without a graceful StopTransaction; session is recovered on the CSMS side via a timeout or explicit cleanup.

## 4 Dependency Map

**Flow 1** 4.1 (Reset Hard) → 4.2 (Boot) → 4.3 → 4.4 → 4.5 → 4.6 → 4.7 → 4.8

**Flow 2** 4.9 (Reset Soft) → 4.10 → 4.11 → 4.12 → 4.13 → 4.14 → 4.15 → 4.16 → 4.17

**Flow 3** 4.18 (Reset Hard) → 4.19 → 4.20 → 4.21